

LegalEase

AI-Powered Legal Document Generator

Phase 2: Requirement Analysis

Project Report

1. Purpose of This Phase

The Requirement Analysis phase converts the ideas and problem statement from the brainstorming phase into a structured, verifiable set of requirements. These requirements guide the system design, development, and testing phases that follow.

2. Stakeholders

Stakeholder	Role / Interest
End Users	Individuals and businesses generating legal documents
Project Team	Designs, develops, tests and maintains the system
Domain/Legal Advisor	Validates clause accuracy and template correctness
Project Guide/Mentor	Reviews progress and academic/technical soundness
Hosting/Cloud Provider	Supplies infrastructure for deployment

3. Requirement Gathering Methods

- Review of existing legal-tech platforms and their feature sets.
- Informal interviews and surveys conducted during the ideation phase.
- Analysis of common legal document structures (rental agreements, NDAs, affidavits).
- Team brainstorming sessions to prioritize a Minimum Viable Product (MVP) feature set.

4. Functional Requirements

ID	Requirement
FR-1	The system shall allow users to register and log in securely.
FR-2	The system shall let users select a document category (e.g., rental agreement, NDA, affidavit).
FR-3	The system shall accept plain-language input describing the user's requirement.
FR-4	The system shall use an AI/NLP model to generate a structured draft document from the input.
FR-5	The system shall allow users to edit the generated document before finalizing.
FR-6	The system shall validate the presence of mandatory clauses for the selected document type.
FR-7	The system shall allow export of the final document as PDF and/or Word format.
FR-8	The system shall maintain a history of previously generated documents per user.

ID	Requirement
FR-9	The system shall support basic e-signature request/collection (stretch goal).

5. Non-Functional Requirements

Category	Requirement
Performance	Document generation should complete within 5–10 seconds for standard templates.
Usability	The interface should require no legal expertise to operate.
Security	User data and generated documents must be encrypted in transit and at rest.
Reliability	The system should be available with at least 99% uptime post-deployment.
Scalability	The architecture should support increasing template categories and user load.
Portability	The web application should work across major browsers and screen sizes.
Maintainability	Codebase should follow modular design for easy addition of new templates.

6. Use Case Summary

- **UC-1 Register/Login:** A new or returning user authenticates to access the platform.
- **UC-2 Select Document Type:** The user browses categories and selects the required document.
- **UC-3 Describe Requirement:** The user enters details in natural language or a guided form.
- **UC-4 Generate Draft:** The AI engine produces a structured draft based on input and templates.
- **UC-5 Edit & Review:** The user reviews, edits and adds/removes clauses as needed.
- **UC-6 Export Document:** The user downloads the finalized document in the desired format.
- **UC-7 View History:** The user accesses previously generated documents from their dashboard.

7. Feasibility Study

7.1 Technical Feasibility

The required technologies — a web front end, a backend API, an NLP/LLM-based generation engine, and a database for templates and user records — are mature, well-documented, and readily available (open-source frameworks and cloud-hosted AI APIs), making the project technically feasible with the current team's skill set.

7.2 Economic Feasibility

Development can proceed using free-tier or low-cost cloud hosting and pay-per-use AI APIs during the MVP stage, keeping initial costs low. Long-term costs relate mainly to AI API usage and hosting scaling with user growth.

7.3 Operational Feasibility

The system is designed to be self-service, requiring minimal training for end users. Internally, it fits within the team's existing development workflow and available tools.

8. Assumptions and Constraints

- The system initially targets a limited set of document categories and one legal jurisdiction.
- AI-generated drafts are treated as first-pass documents; users are advised to seek professional legal review for high-stakes documents.
- Internet connectivity is required, as the AI generation component relies on a hosted model or API.

9. Data Requirements

- User account data: name, email, hashed password, and account creation date.
- Template data: document category, base structure, jurisdiction tag, and associated clause set.
- Session data: raw natural-language input and the structured fields extracted from it.
- Generated document data: final content, associated template and user, creation timestamp, and status.

10. Requirement Traceability Overview

Requirement ID	Traced To
FR-2, FR-3	Use Cases UC-2, UC-3 — template selection and input description
FR-4	Use Case UC-4 — AI-driven draft generation
FR-5	Use Case UC-5 — editing and review
FR-6	Clause Validator design in the Design phase
FR-7	Use Case UC-6 — export functionality
FR-8	Use Case UC-7 — document history

11. Outcome of This Phase

This phase produces a documented set of functional and non-functional requirements, use cases, data requirements, and a feasibility assessment, forming the basis for the system architecture and design decisions made in the next phase.